

Telecom Customer Churn Analysis

1. Project Overview

Customer churn is a critical challenge for telecom companies because losing existing customers directly impacts revenue and increases customer acquisition costs. This project analyzes telecom customer data to identify the major factors contributing to churn and to generate insights that can help improve customer retention strategies.. The goal is to uncover insights into customer retention, identify key drivers of customer churn (such as contract types, payment methods, and demographics), and quantify the revenue lost to churn. These insights will guide strategic business decisions to improve customer loyalty and optimize service offerings.

2. Dataset Summary

- **Rows:** 7,043
- **Columns:** 21
- **Key Features:**
 - **Customer Demographics:** Gender, Senior Citizen status, Partner, Dependents.
 - **Account Details:** Tenure (months), Contract type, Paperless Billing, Payment Method.
 - **Service Subscriptions:** Phone Service, Multiple Lines, Internet Service (DSL/Fiber Optic), Online Security, Tech Support, Streaming capabilities.
 - **Financial Metrics:** Monthly Charges, Total Charges.
- **Target Variable:** Churn (Yes/No) indicating whether the customer left within the last month.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python using Pandas:

- **Data Loading:** Imported the raw dataset (customer_churn.csv).
- **Column Standardization:** Converted all column names to lowercase to ensure consistency and prevent syntax errors during SQL querying.

- **Missing Data Handling:** Identified hidden missing values in the totalcharges column (stored as blank spaces " "). These were replaced with "0", and the column was successfully cast from an object (string) to a float64 data type.
- **Feature Engineering:** Mapped the seniorcitizen binary values (0 and 1) to categorical strings ("No" and "Yes") for better readability in reporting.

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0_7590-WWEG	Female	0	Yes	No	1	No	No phone service	DSL	No	—	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No
1_5575-QWDE	Male	0	No	No	34	Yes	No	DSL	Yes	—	Yes	No	No	One year	No	Mailed check	55.95	1889.5	No
2_3658-QPBY	Male	0	No	No	2	No	No	DSL	Yes	—	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
3_7795-CQOW	Male	0	No	No	45	No	No phone service	DSL	Yes	—	Yes	Yes	No	One year	No	Bank transfer (automatic)	42.30	1940.25	No
4_9337-HOTU	Female	0	No	No	2	Yes	No	Fiber optic	No	—	No	No	No	Month-to-month	Yes	Electronic check	70.70	151.65	Yes
5_8395-CQSC	Female	0	No	No	9	Yes	Yes	Fiber optic	No	—	Yes	No	Yes	Month-to-month	Yes	Electronic check	99.65	820.5	Yes
6_1452-KQVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	—	No	No	Yes	Month-to-month	Yes	Credit card (automatic)	89.10	1949.4	No
7_6713-OKMC	Female	0	No	No	10	No	No phone service	DSL	Yes	—	No	No	No	Month-to-month	No	Mailed check	28.75	301.9	No
8_7892-POOP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	—	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	104.80	3046.05	Yes
9_6388-TABU	Male	0	No	Yes	62	Yes	No	DSL	Yes	—	No	No	No	One year	No	Bank transfer (automatic)	56.15	3487.95	No

- **Database Integration:** Utilized SQLAlchemy to connect the Python environment to a PostgreSQL database, loading the cleaned DataFrame into a table named cus_churn for advanced SQL analysis.

Summary Statistics for Numerical Features:

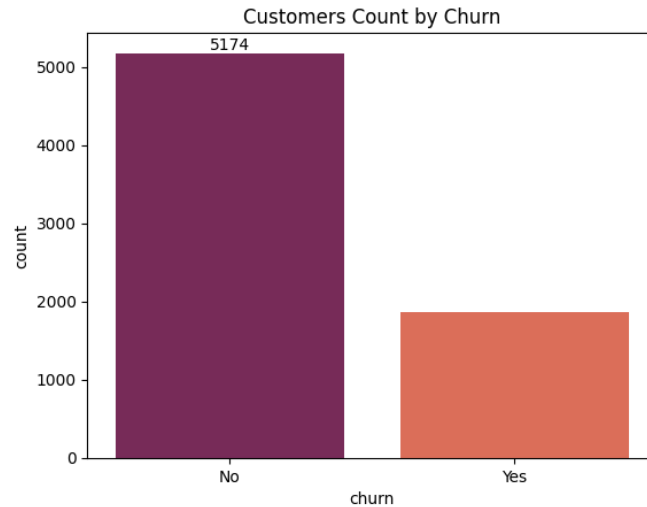
	seniorcitizen	tenure	monthlycharges	totalcharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

4. Visual Representation (Using Matplotlib and Seaborn)

The Target Variable: Churn Distribution

"This visualization represents the balance of our dataset. We used a Seaborn countplot to compare the number of customers who stayed versus those who left."

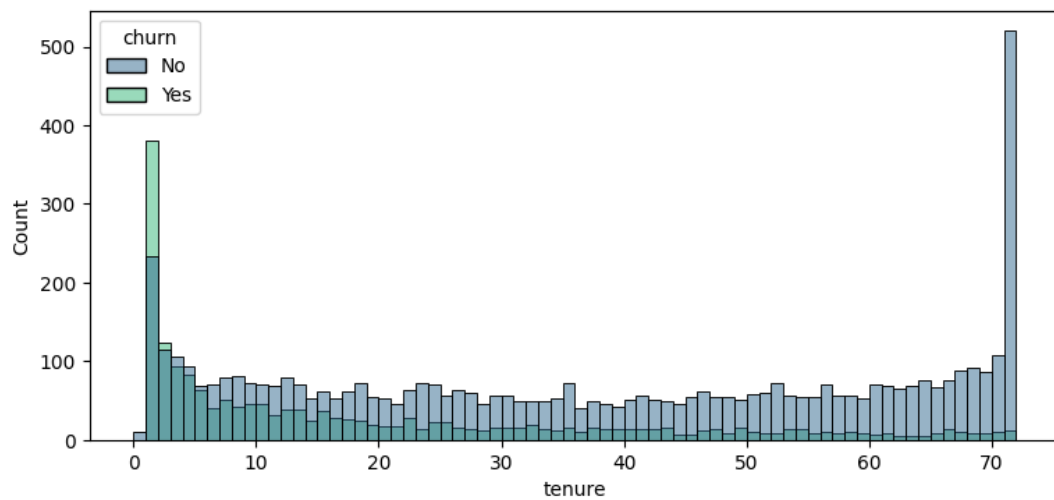
- **Code Tip:** `sns.countplot(x='churn', data=df)`
- **The "Insight":** "The dataset is imbalanced, with approximately 26.5% of customers churned. This baseline helps us understand the scale of customer attrition."



2. The "Danger Zone": Tenure vs. Churn

"To understand *when* customers leave, we used a Matplotlib histogram with a kernel density estimate (KDE). This shows the distribution of customer lifespans."

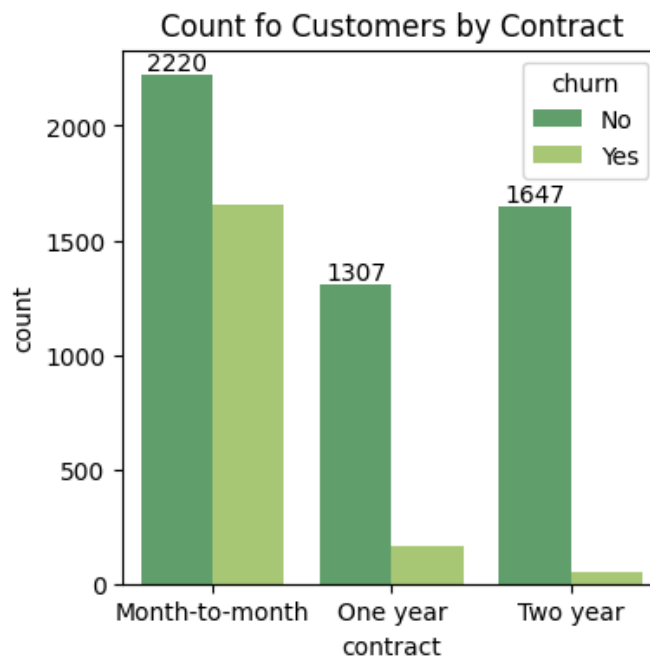
- **Code Tip:** `sns.histplot(x='tenure', hue='churn', multiple='stack', data=df)`
- **The "Insight":** "There is a massive spike in churn during the first 0–5 months of service. If a customer stays past the 20-month mark, their likelihood of leaving drops significantly."



3. Contract Type: The Strongest Predictor

"We used a grouped bar chart to analyze how contract length affects loyalty. This categorical analysis helps identify high-risk account types."

- **Code Tip:** `sns.countplot(x='contract', hue='churn', data=df)`
- **The "Insight":** "Month-to-month customers have the highest churn volume. In contrast, customers on two-year contracts show almost zero attrition, proving that long-term commitments are the most effective retention tool."



4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL using the `cus_churn` table to answer key business questions:

1. Overall Churn Rate

Calculated the baseline percentage of customers leaving the telecom service.

- *Finding:* Over a quarter of the customer base has churned.

	total_customers bigint	churn_rate numeric
1	7043	26.54

2. Churn by Demographics (Senior Citizens)

Compared the churn rate between senior citizens and non-senior customers.

- *Finding:* Senior citizens are highly vulnerable to churn, leaving at nearly double the rate of younger demographics.

	seniorcitizen text	total bigint	churned bigint	churn_rate numeric
1	No	5901	1393	23.61
2	Yes	1142	476	41.68

3. Impact of Contract Type on Retention

Analyzed how commitment length affects a customer's likelihood to leave.

- *Finding:* Month-to-month contracts are the primary driver of churn, whereas 1-year and 2-year contracts show excellent retention.

	contract text	total bigint	churned bigint	churn_rate numeric
1	One year	1473	166	11.27
2	Month-to-mon...	3875	1655	42.71
3	Two year	1695	48	2.83

4. Payment Method Preferences and Churn

Evaluated the stability of different billing methods.

- *Finding:* Customers using Electronic Checks have a significantly higher churn rate compared to those on automated payment systems.

	paymentmethod text	total bigint	churned bigint	churn_rate numeric
1	Electronic check	2365	1071	45.29
2	Credit card (automatic)	1522	232	15.24
3	Mailed check	1612	308	19.11
4	Bank transfer (automat...	1544	258	16.71

5. Average Tenure Comparison

Calculated the average lifespan of retained vs. churned customers.

- *Finding:* Churned customers typically leave before hitting the 18-month mark.

	churn text 	avg_tenure numeric 
1	No	37.57
2	Yes	17.98

6. Internet Service Reliability

Investigated churn behavior based on the type of internet connection provided.

- *Finding:* Fiber optic customers leave at highly elevated rates, suggesting potential pricing dissatisfaction or service instability.

internetservice	churn_rate (%)
Fiber optic	41.89
DSL	18.96
No Internet	7.40

7. Total Revenue Lost to Churn

Quantified the financial impact of customer attrition by summing the totalcharges of churned customers.

metric	financial_impact (USD)
Total Lost Revenue	\$2,862,926.90

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present these insights visually for stakeholders.

- **Top KPI Banners:** Showcasing **7.04K** Total Customers, **\$64.76** Average Monthly Charge, and the critical **26.5%** Churn Rate.
- **Churn by Contract Type (Clustered Column Chart):** Visually highlighting the massive spike in churn volume for "Month-to-month" users compared to flat lines for longer contracts.
- **Churn by Internet Service (Donut Chart):** Demonstrating that Fiber Optic users make up the vast majority of the churn slice.
- **Churn Trend by Tenure (Area Chart):** Displaying a steep drop-off curve in the first 0–12 months, which slowly flattens out as customers reach year 2 and beyond.
- **Interactive Slicers:** Added dropdowns for Gender and Senior Citizen status to allow marketing teams to filter the entire dashboard dynamically.



6. Business Recommendations

Based on the SQL findings and Python data visualizations, the following strategic actions are recommended:

- **Incentivize Contract Upgrades:** With Month-to-month contracts driving 42.7% churn, marketing should launch targeted campaigns offering discounted hardware or streaming bundles to users willing to sign 1-year commitments.
- **Migrate Users to Auto-Pay:** Customers paying via "Electronic check" are churning at an alarming 45.2%. The business should offer a \$5/month statement credit to users who switch to "Credit card (automatic)" or "Bank transfer", which boast churn rates below 17%.
- **Investigate Fiber Optic Service:** A 41.8% churn rate in Fiber Optic suggests an issue with the product itself. Technical support tickets and pricing tiers for Fiber users should be audited immediately to ensure competitive value and network reliability.
- **Focus on the "First 18 Months":** Data shows churned users leave on average at 17.9 months. Customer Success teams should implement proactive check-ins at month 3, 6, and 12 to resolve grievances before the customer decides to cancel.
- **Tailored Senior Support:** Senior citizens are churning at 41.6%. Creating simplified billing statements and dedicated "priority" customer service routing for seniors could drastically improve their retention.